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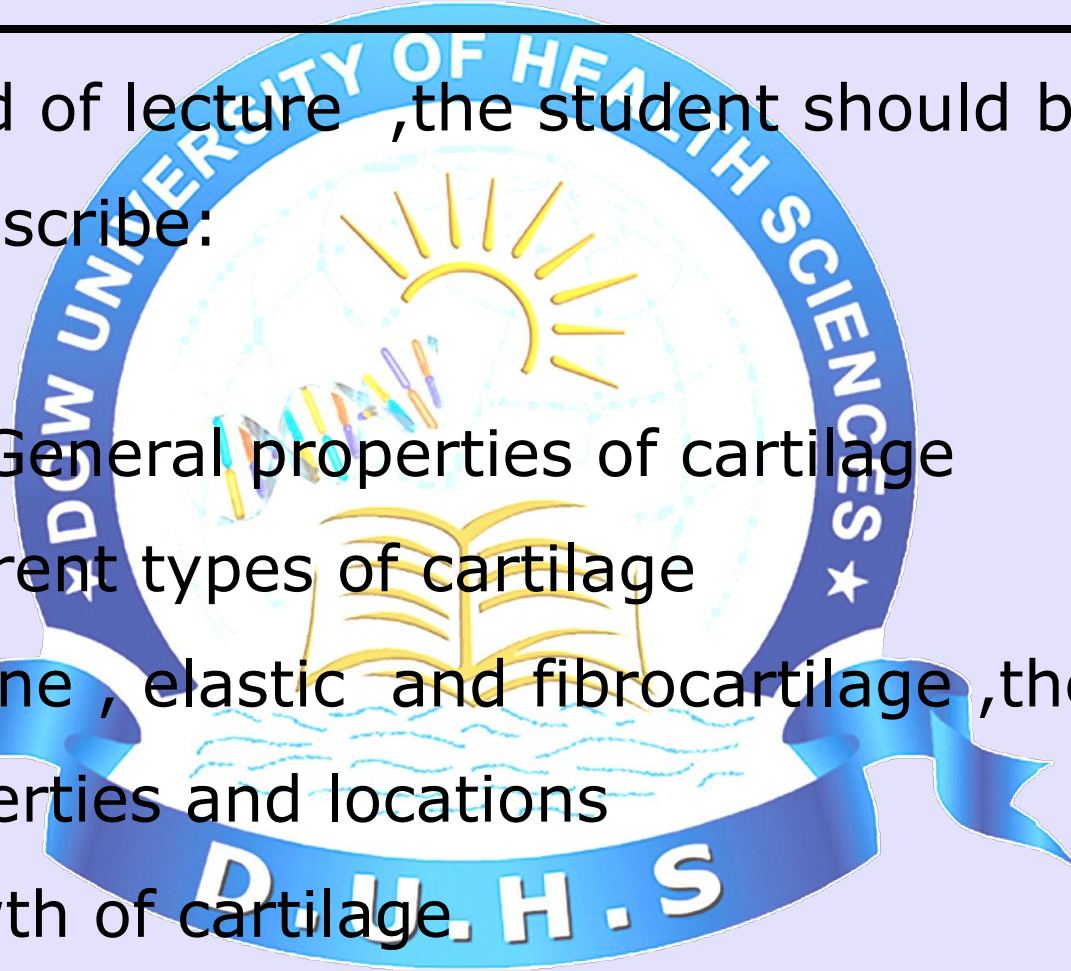


CLASSIFICATION & HISTOLOGY OF CARTILAGE

LEARNING OBJECTIVES

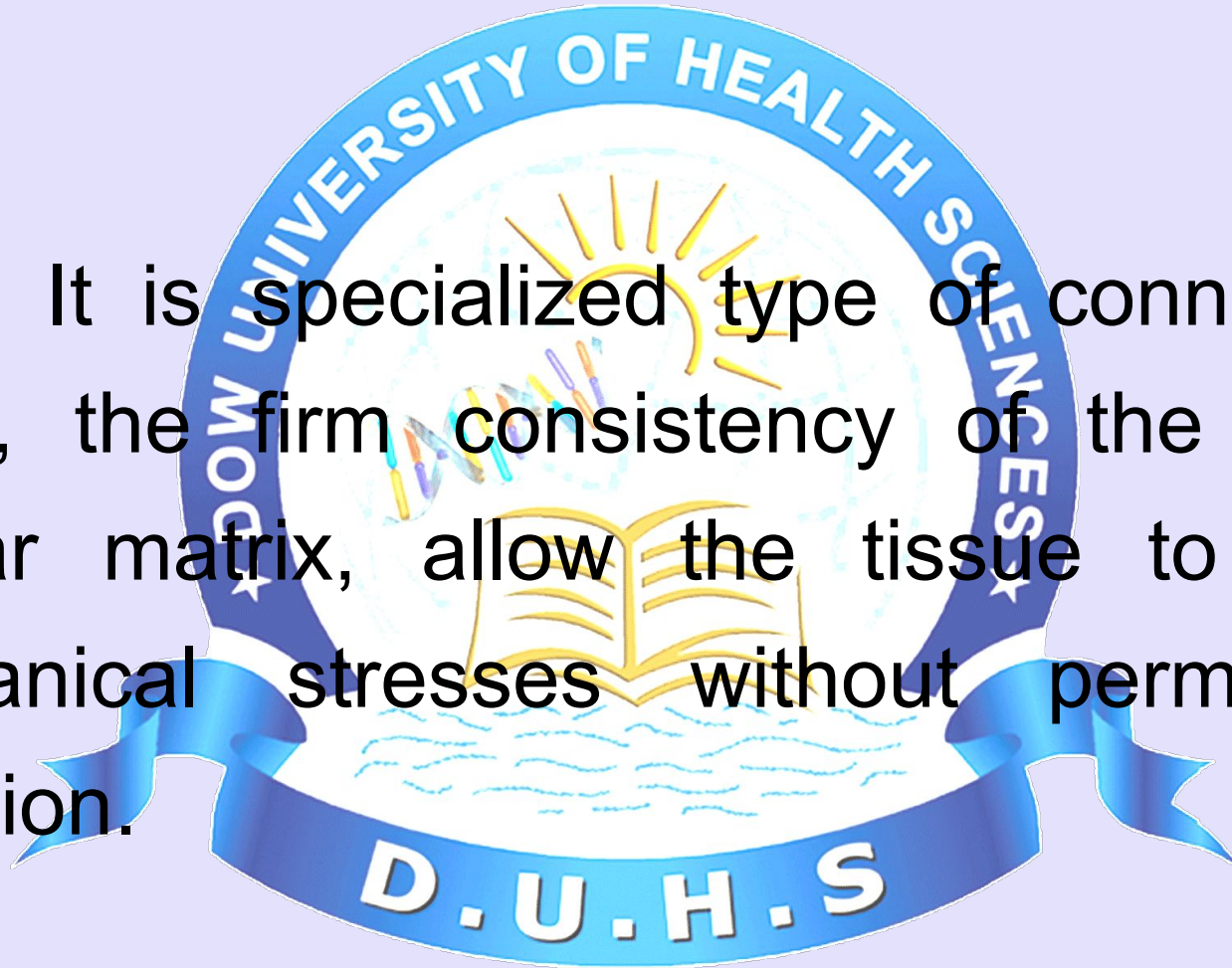
At the end of lecture ,the student should be able to describe:

- The General properties of cartilage
- Different types of cartilage
- Hyaline , elastic and fibrocartilage ,their properties and locations
- Growth of cartilage



CARTILAGE

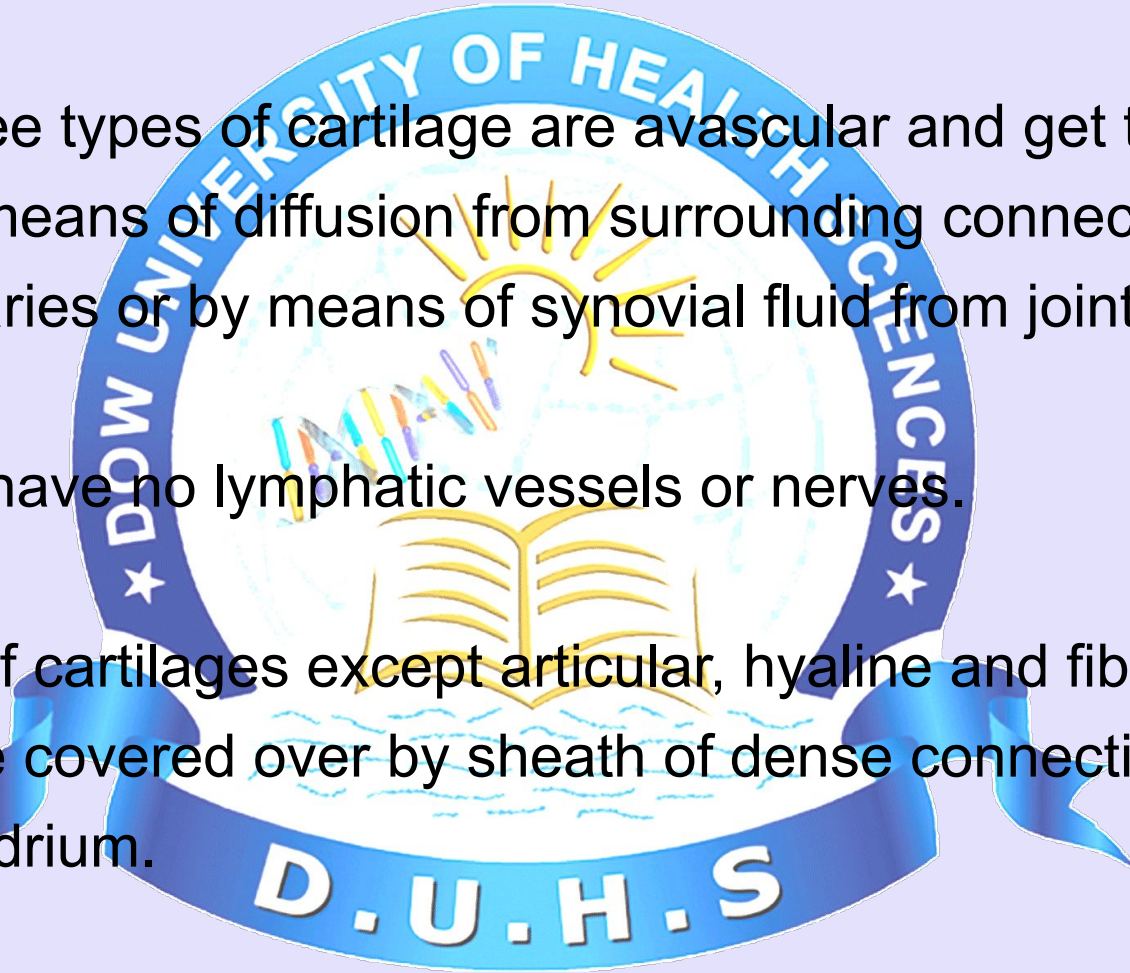
It is specialized type of connective tissue, the firm consistency of the extra cellular matrix, allow the tissue to bear mechanical stresses without permanent distortion.



GENERAL PROPERTIES OF CARTILAGE



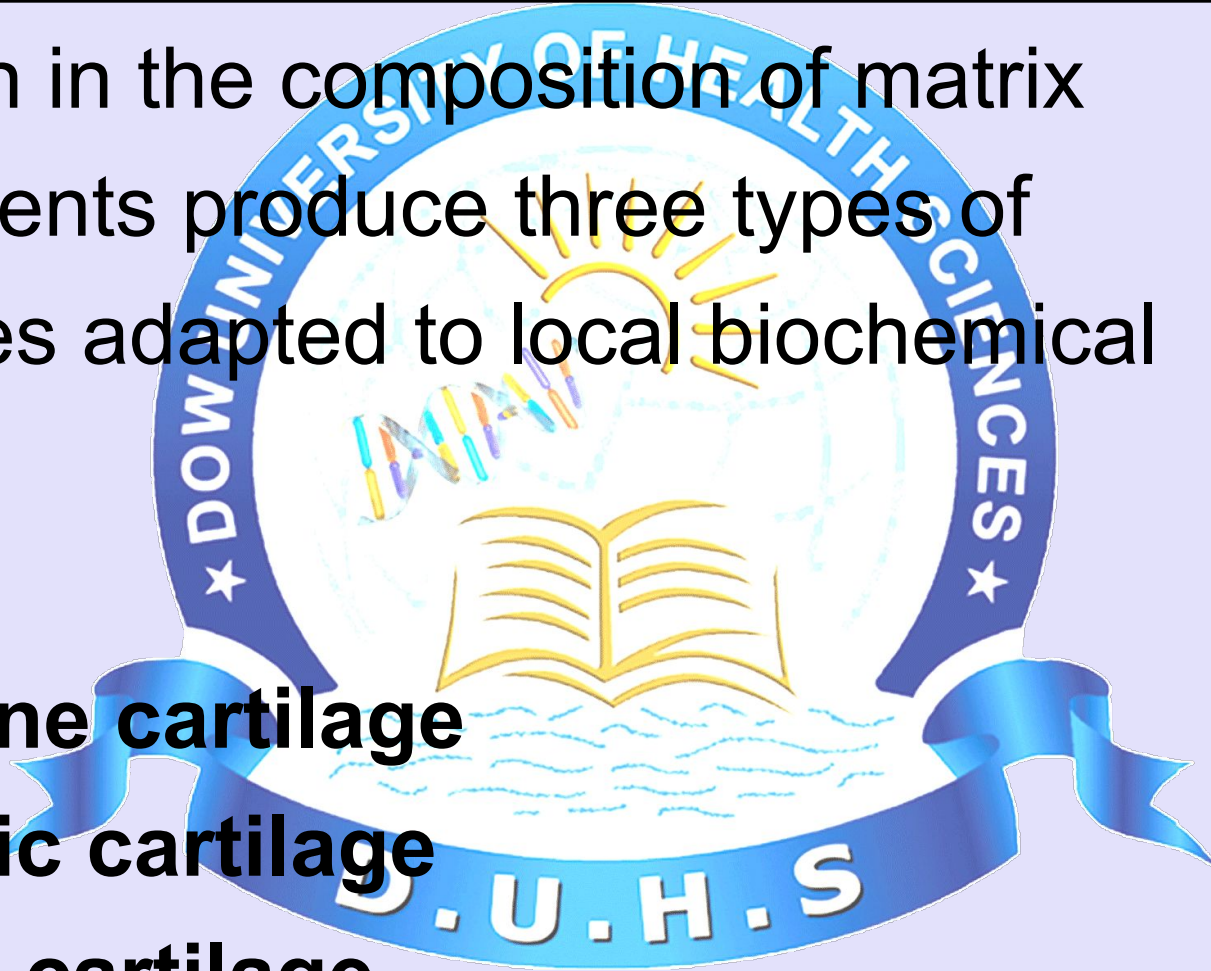
1. All the three types of cartilage are avascular and get their nutrition by means of diffusion from surrounding connective tissue capillaries or by means of synovial fluid from joint cavities.
2. Cartilage have no lymphatic vessels or nerves.
3. All types of cartilages except articular, hyaline and fibro cartilage, are covered over by sheath of dense connective tissue the perichondrium.



TYPES OF CARTILAGE

Variation in the composition of matrix components produce three types of cartilages adapted to local biochemical needs.

- 1. Hyaline cartilage**
- 2. Elastic cartilage**
- 3. Fibro cartilage**



COMPONENTS OF CARTILAGE



As for other types of connective tissue it has the three components:

1. Cartilage Cell

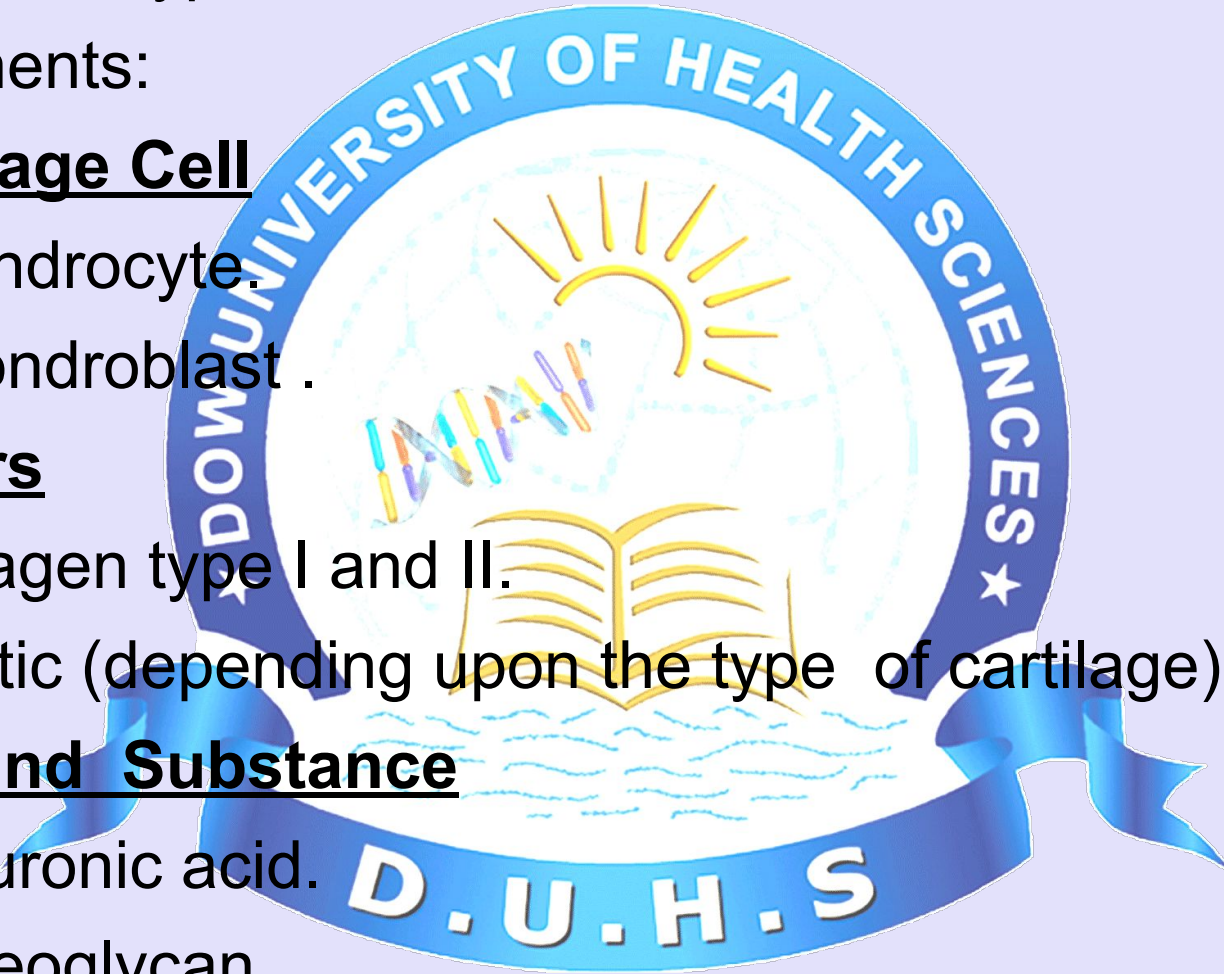
- a) Chondrocyte.
- b) Chondroblast .

2. Fibers

- a) Collagen type I and II.
- b) Elastic (depending upon the type of cartilage).

3. Ground Substance

- a) Hyaluronic acid.
- b) Proteoglycan.
- c) Glycoprotein.



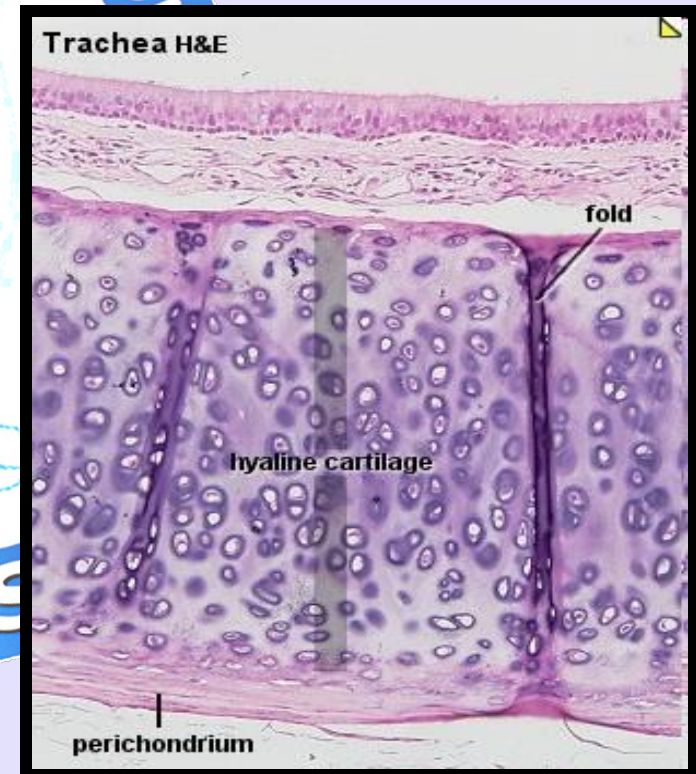
HYALINE CARTILAGE

The most abundant among the three types.

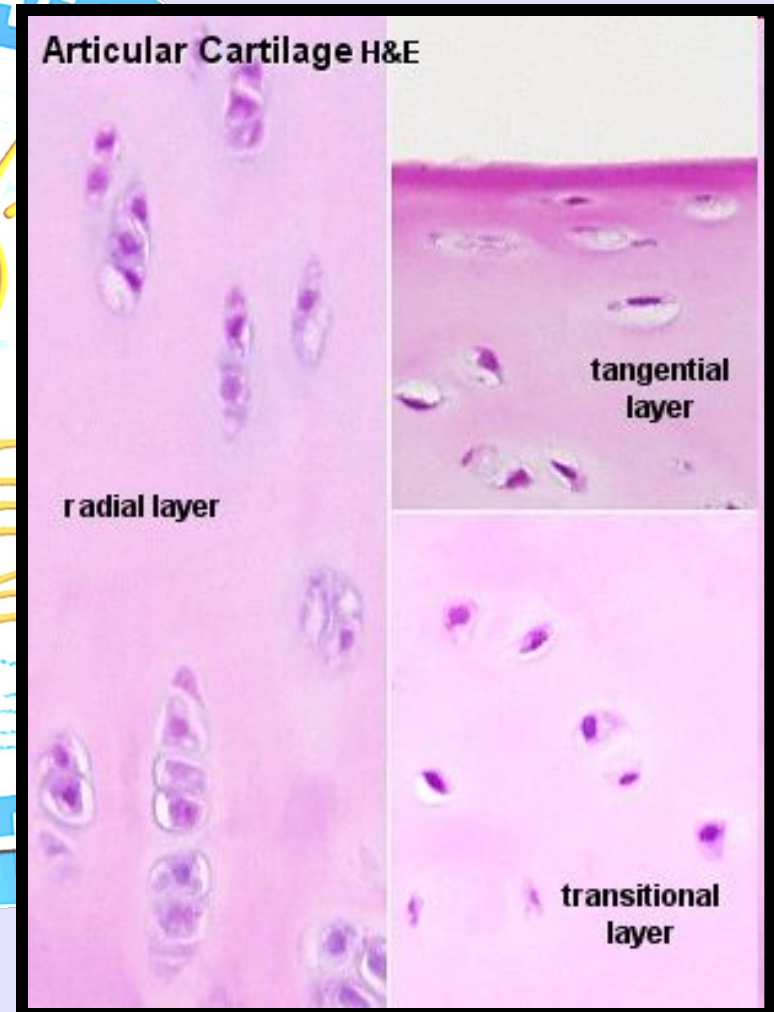
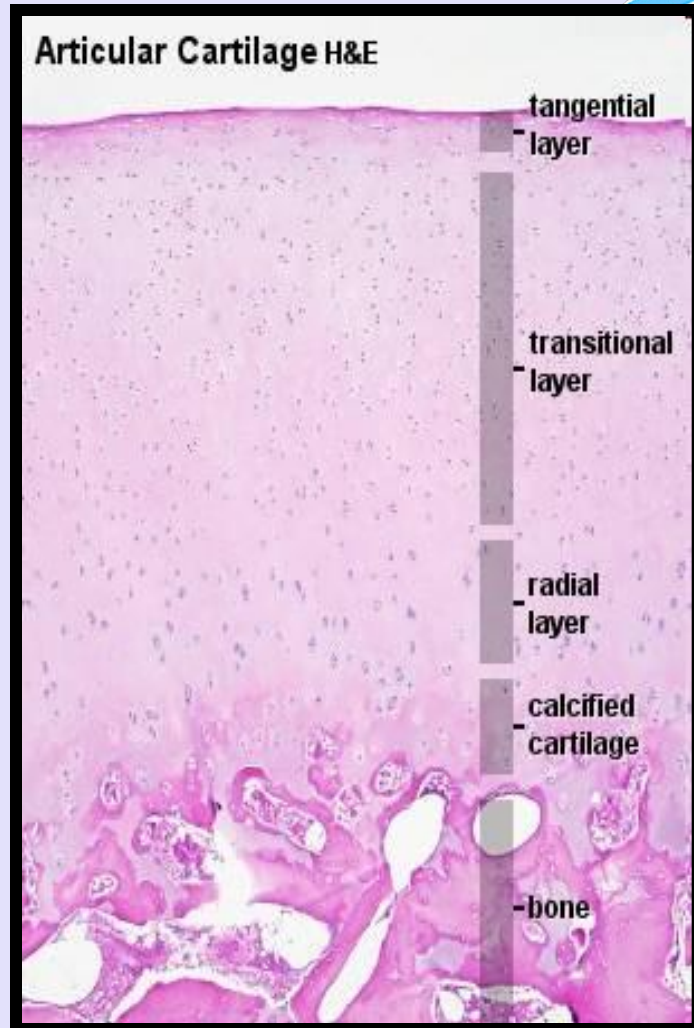
Appearance: Bluish-white when fresh.

Sites:

- Temporary embryonic skeleton
- Epiphyseal plate
- Articular surfaces of various synovial joints.
- In the wall of large respiratory passages.



ARTICULAR CARTILAGE-AN EXAMPLE OF HYALINE CARTILAGE

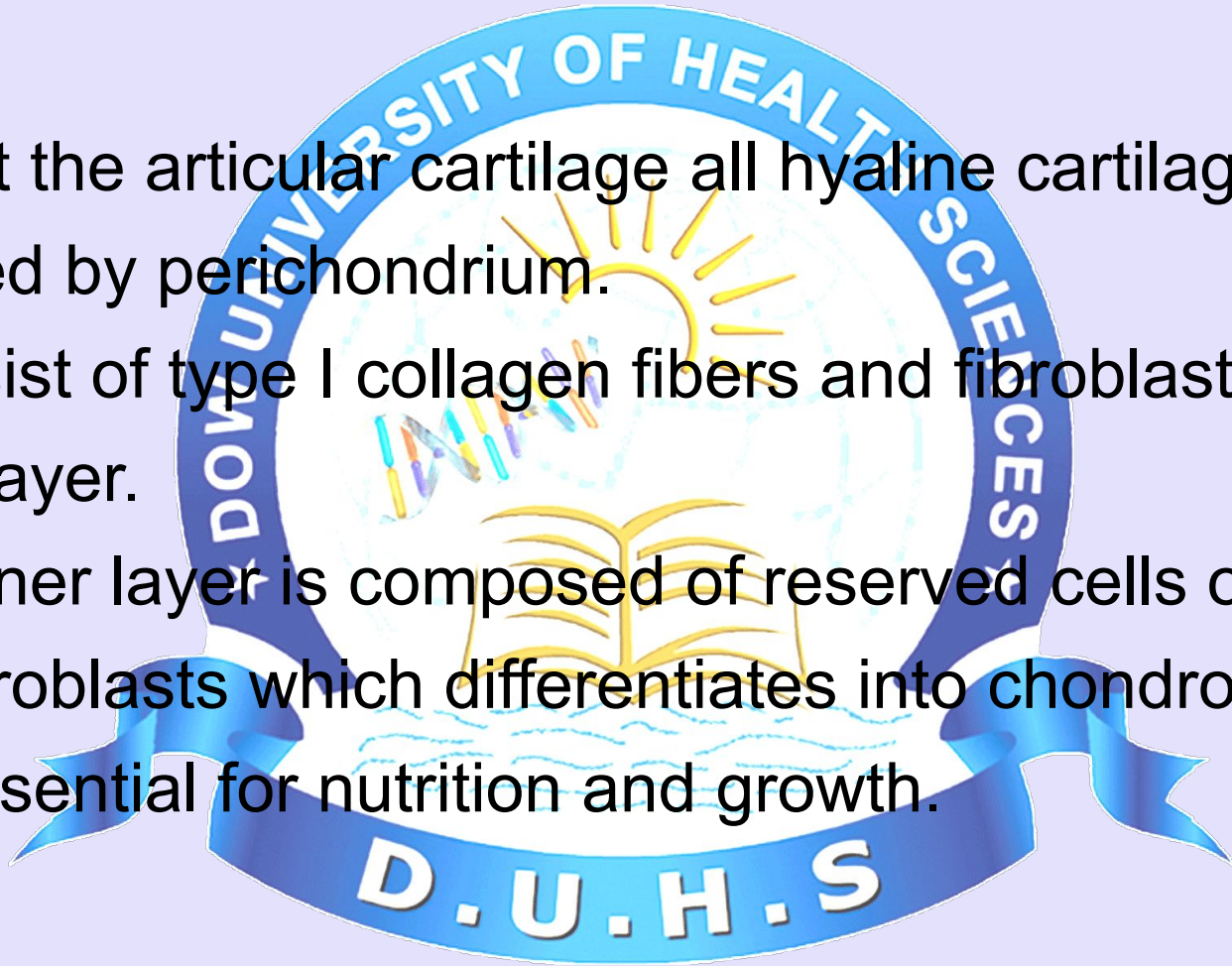


PERICHONDRIUM

Except the articular cartilage all hyaline cartilage is covered by perichondrium.

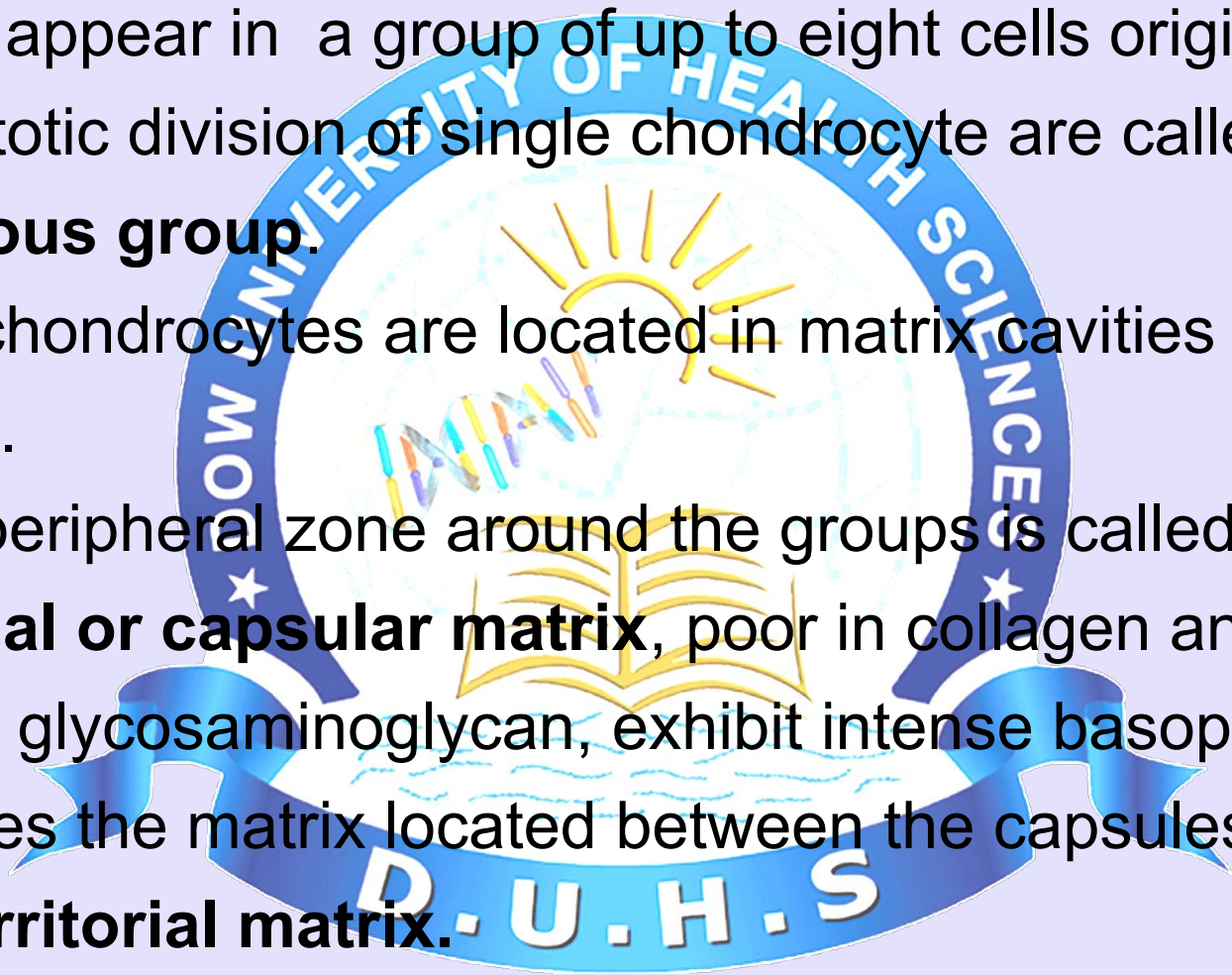
It consists of type I collagen fibers and fibroblasts in the outer layer.

The inner layer is composed of reserved cells or chondroblasts which differentiate into chondrocytes. It is essential for nutrition and growth.

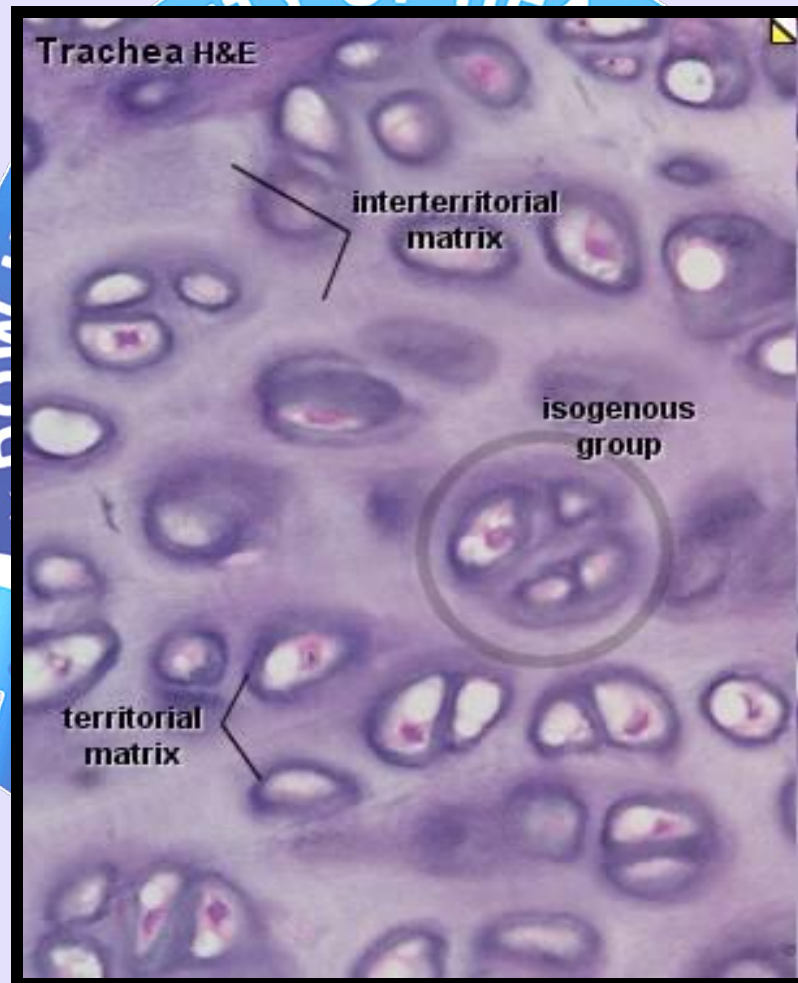


CHONDROCYTE

1. They appear in a group of up to eight cells originating from mitotic division of single chondrocyte are called **isogenous group**.
2. The chondrocytes are located in matrix cavities called lacunae.
3. The peripheral zone around the groups is called **territorial or capsular matrix**, poor in collagen and rich in sulfated glycosaminoglycan, exhibit intense basophilia than does the matrix located between the capsules called **inter territorial matrix**.



CHONDROCYTE



MATRIX

Fibers + ground substance

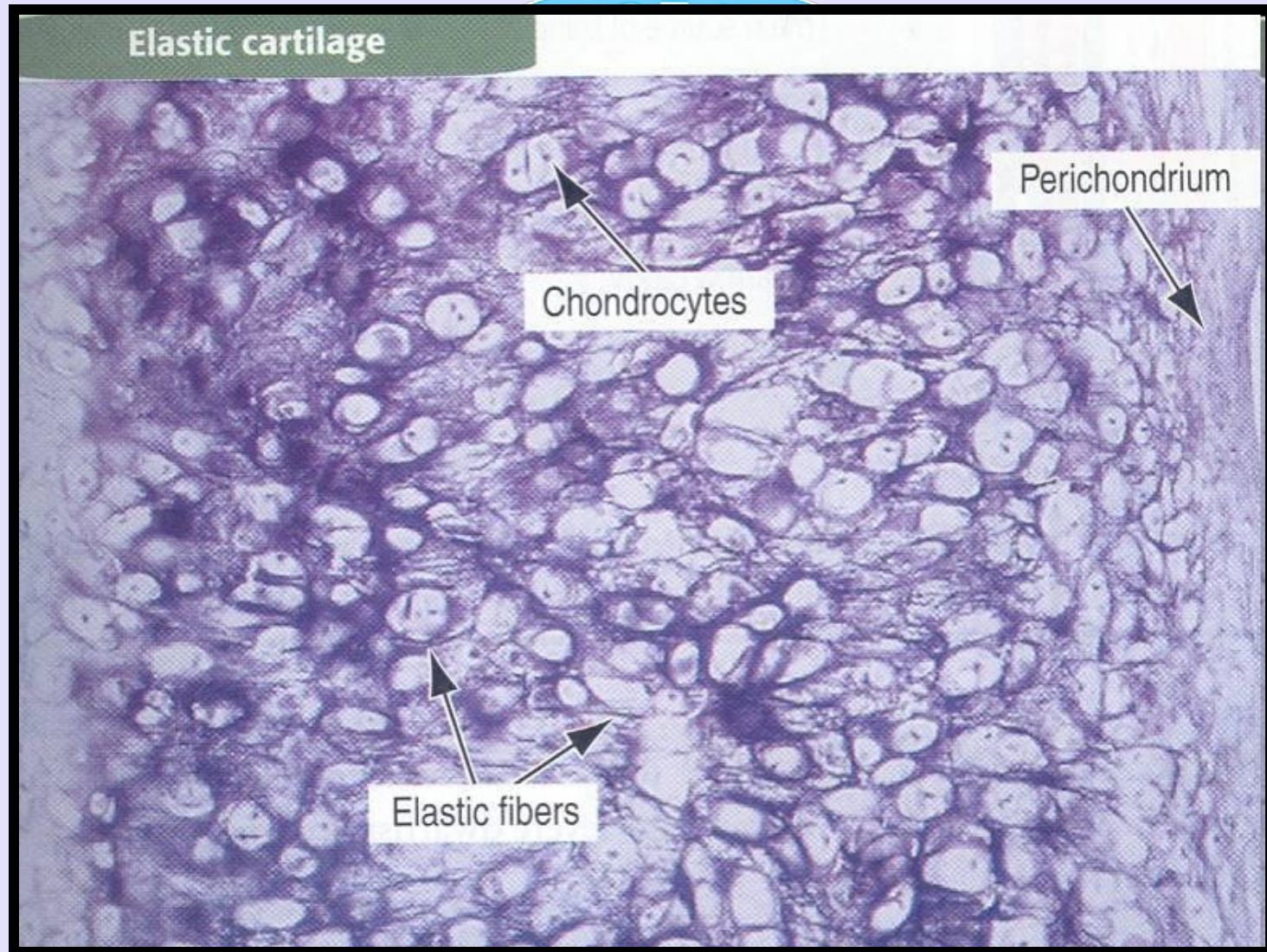
1. It has primarily type II collagen which is in the form of fibrils and is indiscernible from the ground substance.
2. The ground substance contains chondroitin 4 and 6 sulfate, and keratan sulfate in the proteoglycans.
3. The hyaluronic acid binds to these proteoglycans and is responsible for high content of solvation water.
4. Chondronectin is the glycoprotein which adheres chondrocytes to matrix collagen.

ELASTIC CARTILAGE

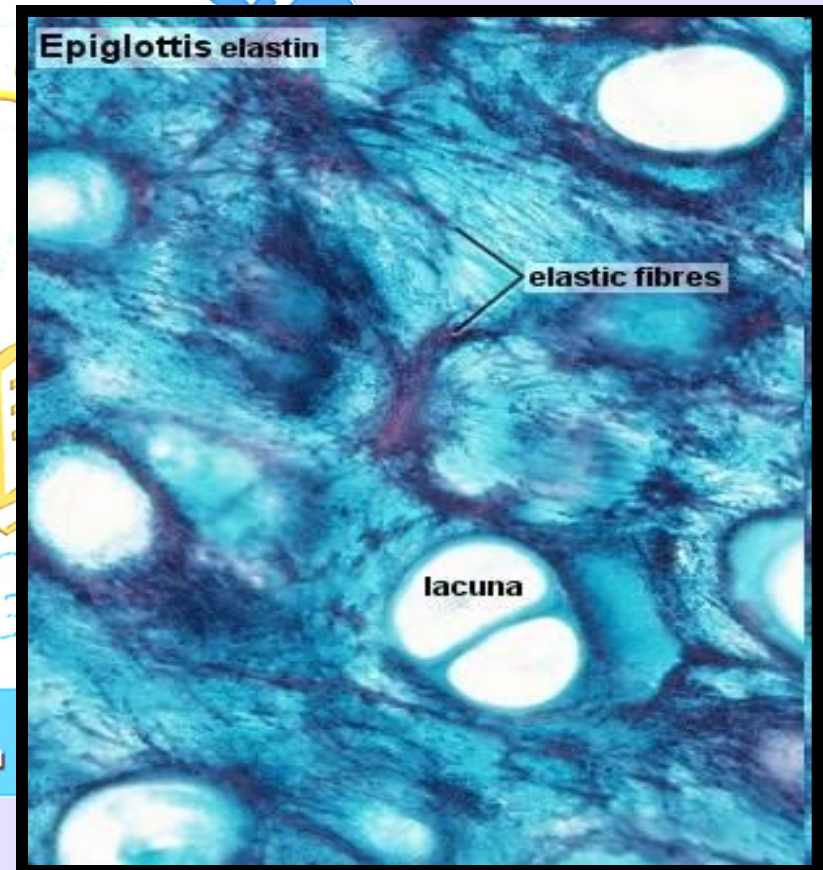
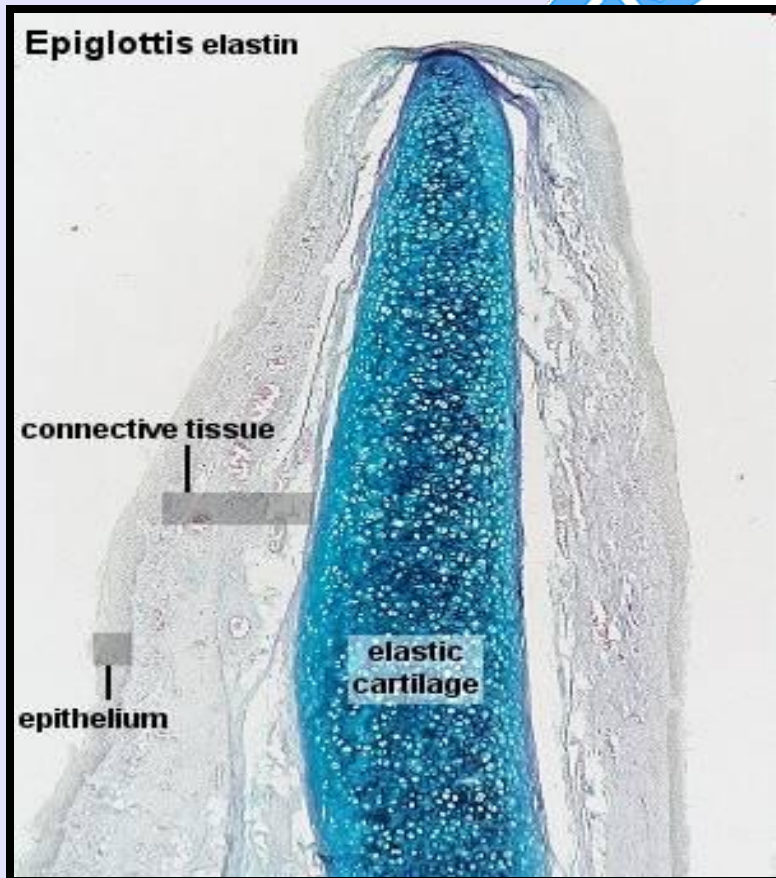
1. Fresh elastic cartilage has yellowish color because of the presence of elastin.
2. It is identical to hyaline cartilage but in addition to collagen type II it contains an abundant network of fine elastic fibers.
3. It also possess perichondrium.

Sites: Auricle of the ear, wall of external auditory canal , auditory tube and epiglottis.

ELASTIC CARTILAGE



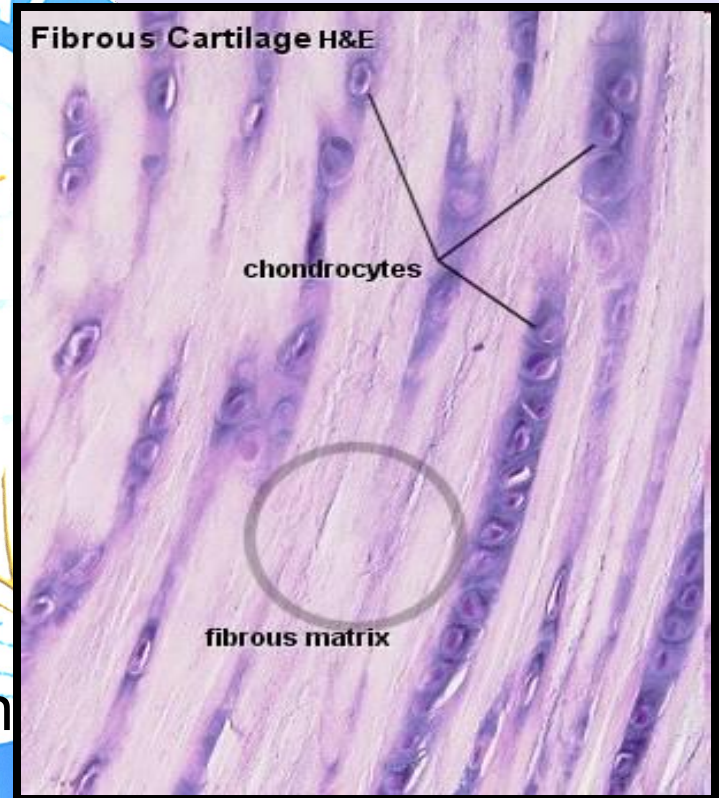
ELASTIC CARTILAGE



FIBROCARILAGE



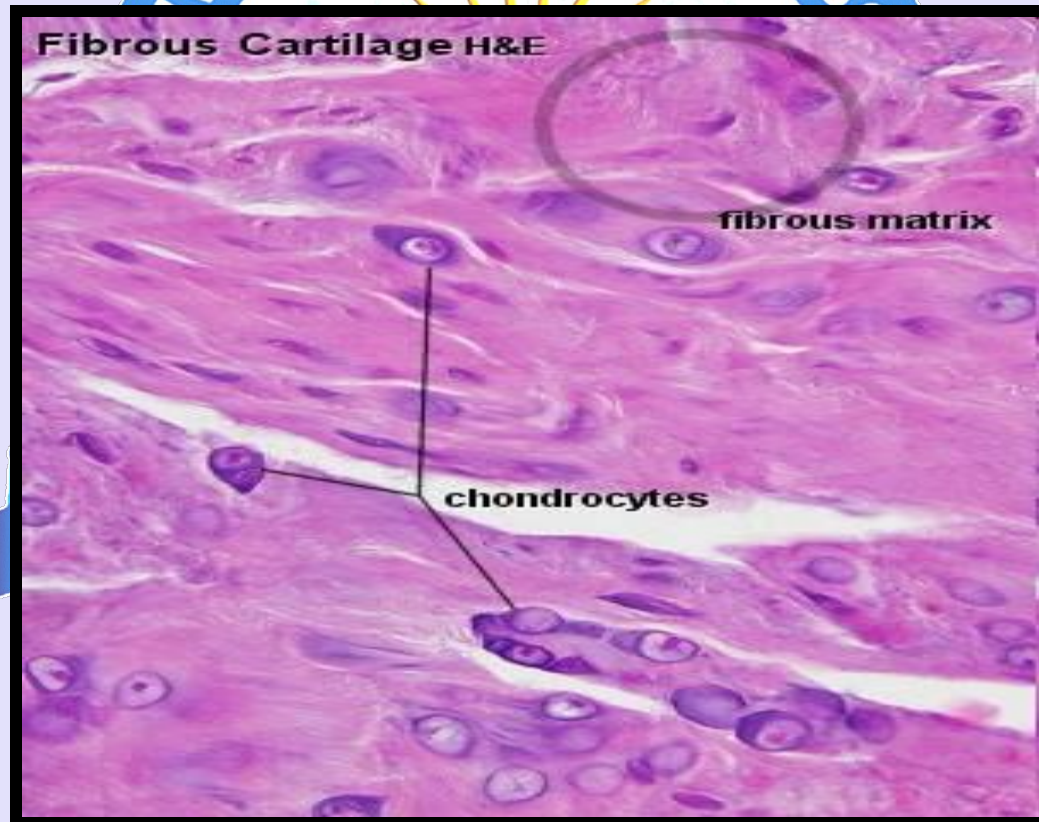
1. It has characteristics intermediate between those of dense connective tissue and hyaline cartilage.
2. Always associated with dense connective tissue and the border area between the two tissues is not clear cut.
3. Chondrocytes appear singly or in isogenous groups arranged in rows.
4. Matrix is acidophilic because it contains a large amount of collagen type I.
5. No identifiable perichondrium.



FIBROCARILAGE

SITE:

Intervertebral Disk , symphysis pubis, glenoid fossa, acetabulum of hip, inter articular joints of the sternum and the clavicle.



GROWTH OF CARTILAGE



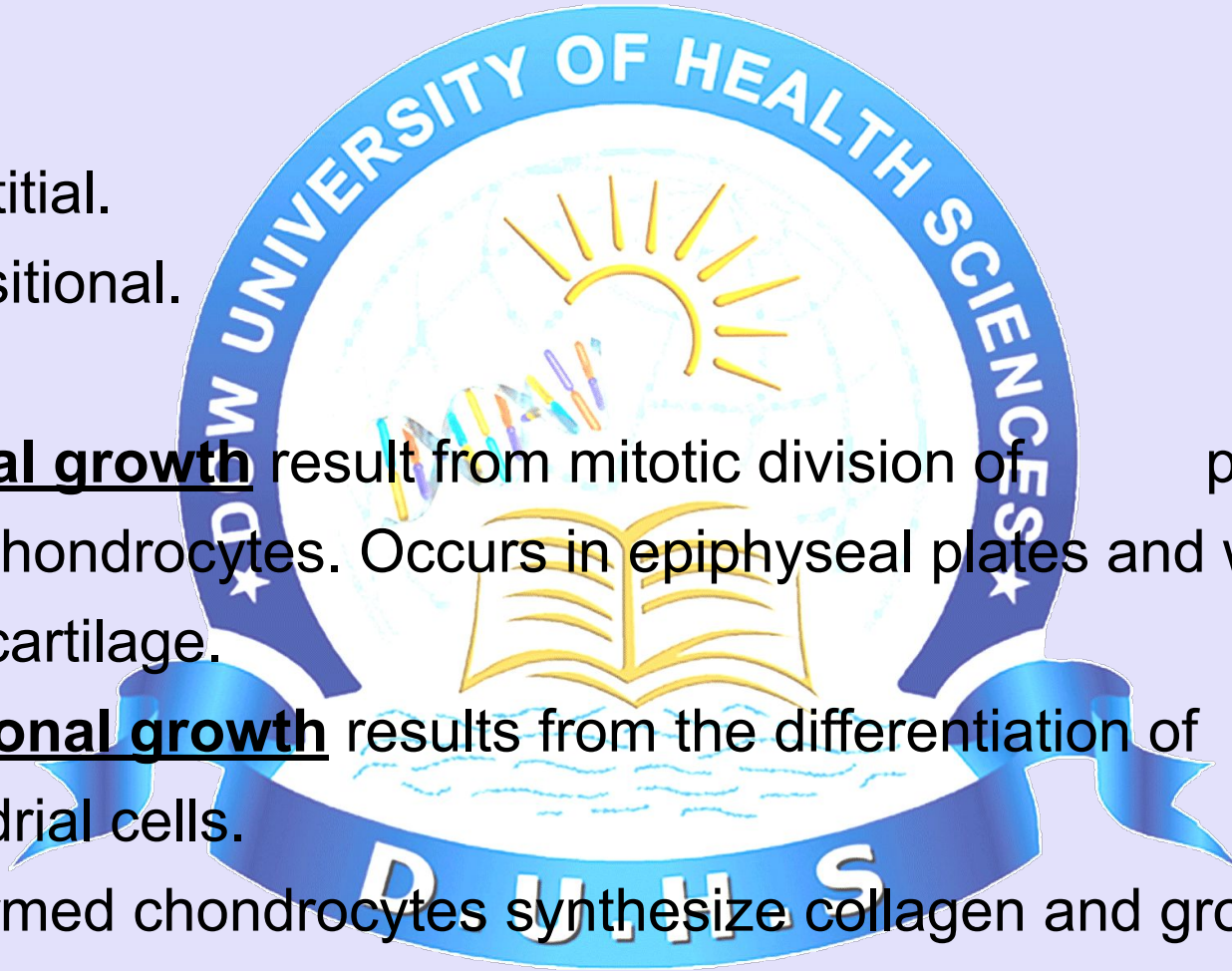
2 Types

- A) Interstitial.
- B) Appositional.

Interstitial growth result from mitotic division of pre existing chondrocytes. Occurs in epiphyseal plates and within articular cartilage.

Appositional growth results from the differentiation of perichondrial cells.

Newly formed chondrocytes synthesize collagen and ground substance.



REFERENCES

- MEDICAL HISTOLOGY BY LIAQ HUSSAIN SIDDIQUI
- BASIC HISTOLOGY BY JUNQUEIRA

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